

UNIT- VI

UNIT – VI ANALYTICAL LEARNING

- Learning with perfect domain theory (Book-2)
 - Explanation based Learning
 - Inductive analytical approach to learning
 - KBANN algorithm

EXPLANATION BASED LEARNING

- Inductive learning methods such as Neural Network and Decision Tree Learning, Inductive Logic Programming, and Genetic Algorithms require a certain number of training examples to achieve a given level of generalization accuracy.
- Analytical learning uses Prior Knowledge and Deductive Reasoning to augment the information provided by the training examples, so that it is not subject to these same bounds.
- EX: Explanation-Based Learning (EBL) when the learner's prior knowledge is correct and complete.
- EBL is an analytical approach which uses prior knowledge
- Prior knowledge is used to analyze, or explain, how each observed training example satisfies the target concept.
- Explanation is used to distinguish the relevant features vs irrelevant, and so examples can be generalized based on logical rather than statistical reasoning.

- EBL algorithms accept **explicit prior knowledge** as an input, in **addition to the input training data**.
- This generalizes more accurately than inductive systems that rely on the data alone.
- It uses **prior knowledge to reduce the complexity of the hypothesis space to be searched**.
- Consider the task of learning to play chess. (Figure 11.1)
- Chess program to learn to recognize important classes of game positions, such as the target concept "chessboard positions in which black will lose its queen within two moves" .
- The chessboard is fairly complex (32 pieces and 64 squares), and because the particular patterns that capture this concept are fairly subtle.
- Provide thousands of training examples like Fig.11.1 to expect an inductively learned hypothesis to generalize correctly to new situations.

- The answer appears to be that people rely heavily on explaining, or analyzing, the training example in terms of their prior knowledge about the legal moves of chess.

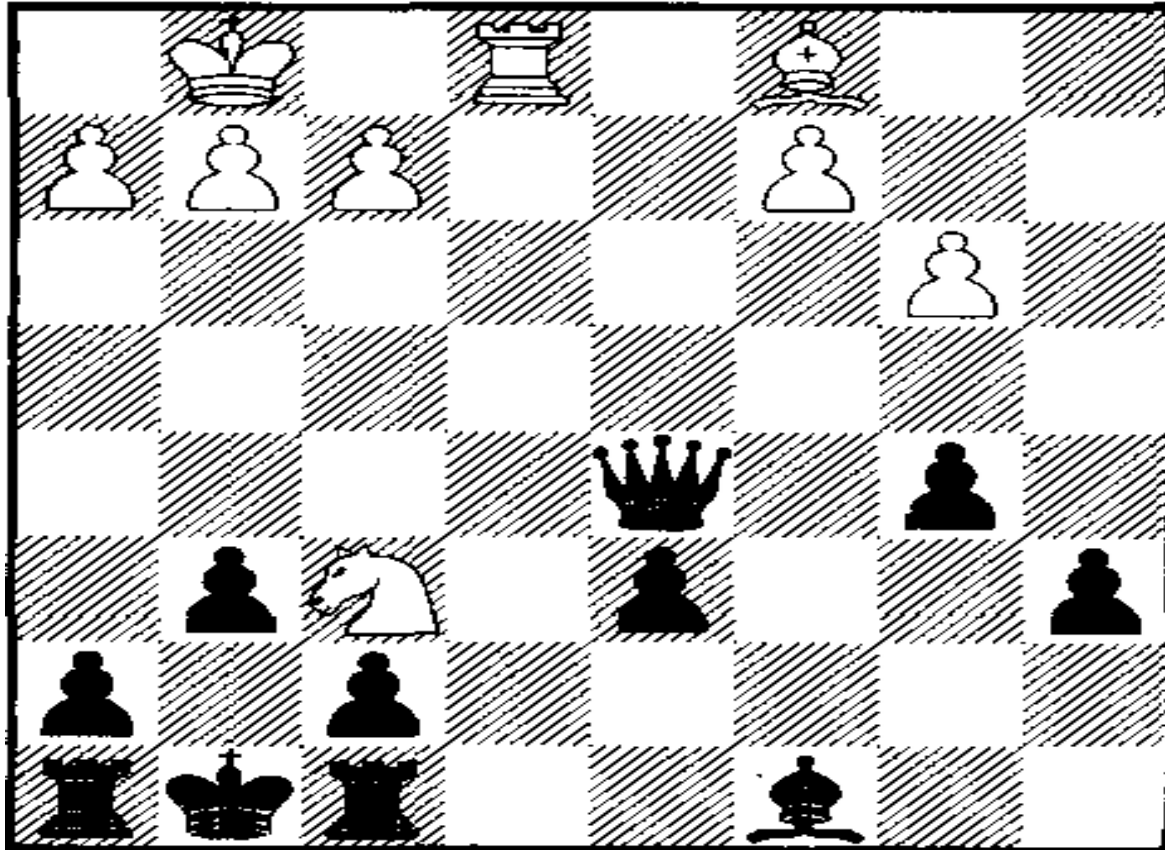


FIGURE 11.1 : A positive example of the target concept "chess positions in which black will lose its queen within two moves." Note the white knight is simultaneously attacking both the black king and queen. Black must therefore move its king, enabling white to capture its queen.

The Inductive Generalization Problem

Given:

- Instances
- Hypotheses
- Target Concept
- Training examples of target concept

Determine:

- Hypotheses consistent with the training examples

The Analytical Generalization Problem

Given:

- Instances
- Hypotheses
- Target Concept
- Training examples of target concept
- *Domain theory for explaining examples*

Determine:

- Hypotheses consistent with the training examples *and the domain theory*

An Analytical Generalization Problem

Given:

- Instances: pairs of objects
- Hypotheses: sets of horn clause rules
- Target Concept: Safe-to-stack(x,y)
- Training Example: Safe-to-stack(OBJ1,OBJ2)

```
On(OBJ1,OBJ2)
Isa(OBJ1,BOX)
Isa(OBJ2,ENDTABLE)
Color(OBJ1,RED)
Color(OBJ2,BLUE)
Volume(OBJ1,.1)
Density(OBJ1,.1)
...
```

- Domain Theory:

```
Safe-To-Stack(x,y) :- Not(Fragile(y))
Safe-To-Stack(x,y) :- Lighter(x,y)
Lighter(x,y) :- Weight(x,wx), Weight(y,wy),
                Less(wx,wy)
Weight(x,w) :- Volume(x,v), Density(x,d),
               Equal(w, v*d)
Weight(x,5) :- Isa(x, ENDTABLE)
...
```

Determine:

- Hypotheses consistent with training examples and domain theory

LEARNING WITH PERFECT DOMAIN THEORIES: PROLOG-EBG

- PROLOG is logic programming language
- PROLOG is a declarative language
- Logic is expressed in the form of relations (facts and rules)
- Consider EBL from domain theories that are perfect, that is, domain theories that are correct and complete.
- A domain theory is said to be **correct** if each of its assertions is a truthful statement about the world.
- A domain theory is said to be **complete** with respect to a given target concept and instance space, if the domain theory covers every positive example in the instance space.
- If we follow the usual PROLOG convention that un-provable assertions are assumed to be false, then this definition of completeness includes full coverage of both positive and negative examples by the domain theory.
- First, there are cases in which it is feasible to provide a perfect domain theory.
- Second, in many other cases it is unreasonable to assume that a perfect domain theory is available.

Learning from Perfect Domain Theories

Assumes domain theory is *correct* (error-free)

- Prolog-EBG is algorithm that works under this assumption
- This assumption holds in chess and other search problems
- Allows us to assume explanation = proof
- Later we'll discuss methods that assume *approximate* domain theories

Prolog EBG

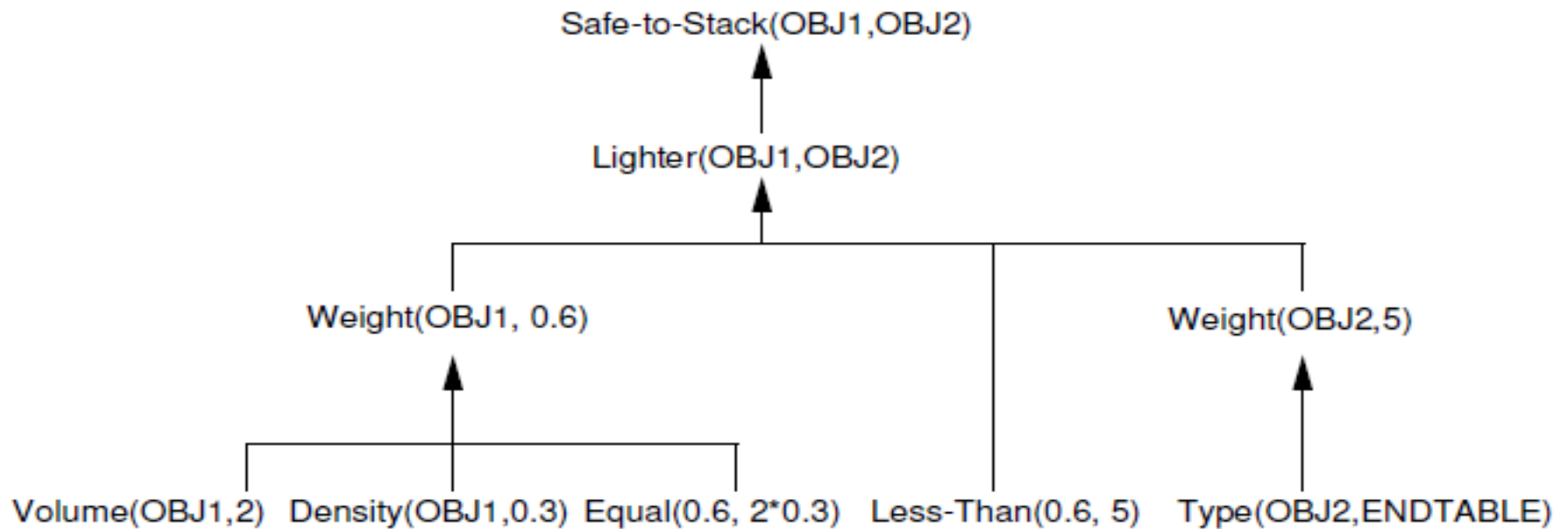
Initialize hypothesis = {}

For each positive training example not covered by hypothesis

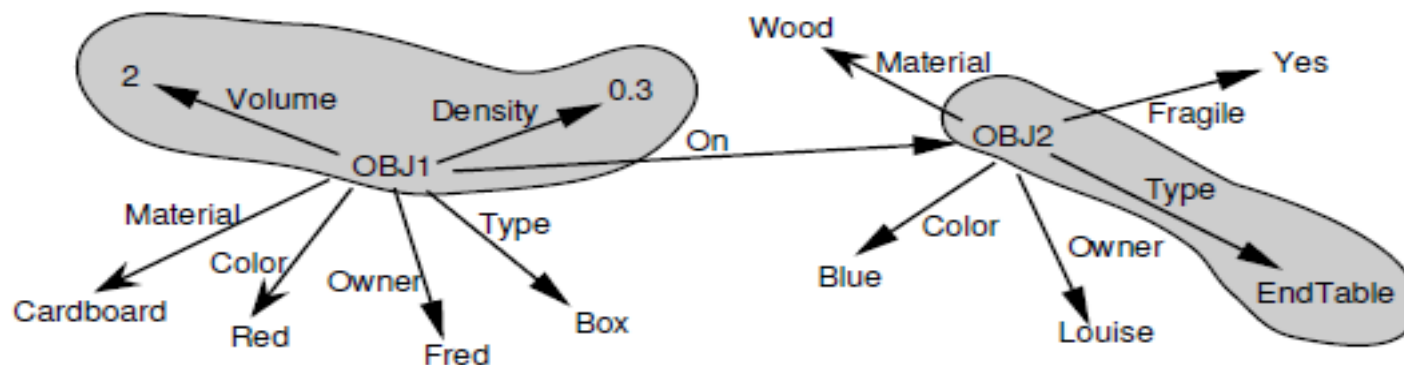
1. **Explain** how training example satisfies target concept, in terms of domain theory
2. **Analyze** the explanation to determine the most general conditions under which this explanation (proof) holds
3. **Refine** the hypothesis by adding a new rule, whose preconditions are the above conditions, and whose consequent asserts the target concept

Explanation of a Training Example

Explanation:



Training Example:



REMARKS ON EXPLANATION-BASED LEARNING

- The following are the key properties of this algorithm.
- It produces justified general hypotheses by using prior knowledge to analyze individual examples.
- The explanation of how the example satisfies the target concept determines which example attributes are relevant.
- Analysis of the explanation allows deriving more general constraints on the values of the relevant features.
- Each learned Horn clause corresponds to a sufficient condition for satisfying the target concept.
- The generality depends on the formulation of the domain theory and on the sequence.
- It assumes that the domain theory is correct and complete.

Perspectives on Prolog-EBG

- Theory-guided generalization from examples
- Example-guided operationalization of theories
- "Just" restating what learner already "knows"

Is it learning?

- Are you learning when you get better over time at chess?
 - Even though you already know everything in principle, once you know rules of the game...
- Are you learning when you sit in a mathematics class?
 - Even though those theorems follow deductively from the axioms you've already learned...

Inductive

Inductive and Analytical Learning

Inductive learning

Hypothesis fits data

Statistical inference

Requires little prior knowledge

Syntactic inductive bias

Analytical learning

Hypothesis fits domain the

Deductive inference

Learns from scarce data

Bias is domain theory

Inductive learning

Analytical learning

Plentiful data

No prior knowledge

Perfect prior knowledge

Scarce data

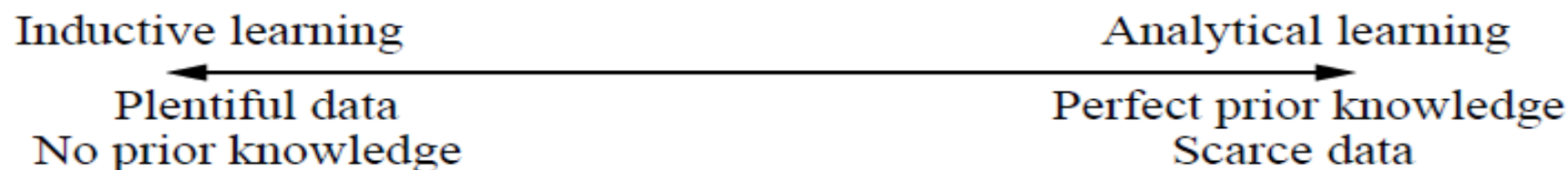
Combining Inductive and Analytical Learning

[Read Ch. 12]

[Suggested exercises: 12.1, 12.2, 12.6, 12.7, 12.8]

- Why combine inductive and analytical learning?
- KBANN: Prior knowledge to initialize the hypothesis
- TangetProp, EBNN: Prior knowledge alters search objective
- FOCL: Prior knowledge alters search operators

What We Would Like



General purpose learning method:

- No domain theory $\rightarrow$ learn as well as inductive methods
- Perfect domain theory $\rightarrow$ learn as well as PROLOG-EBG
- Accomodate arbitrary and unknown errors in domain theory
- Accomodate arbitrary and unknown errors in training data

Domain Theory

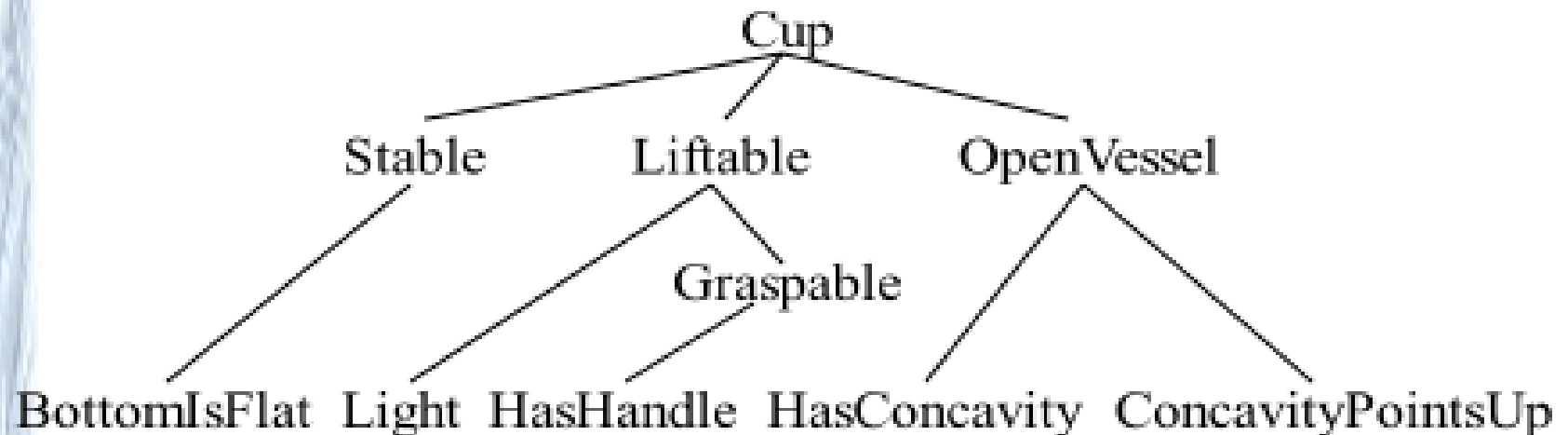
Cup $\leftarrow$ Stable, Lifiable, OpenVessel

Stable $\leftarrow$ BottomIsFlat

Lifiable $\leftarrow$ Graspable, Light

Graspable $\leftarrow$ HasHandle

OpenVessel $\leftarrow$ HasConcavity, ConcavityPointsUp

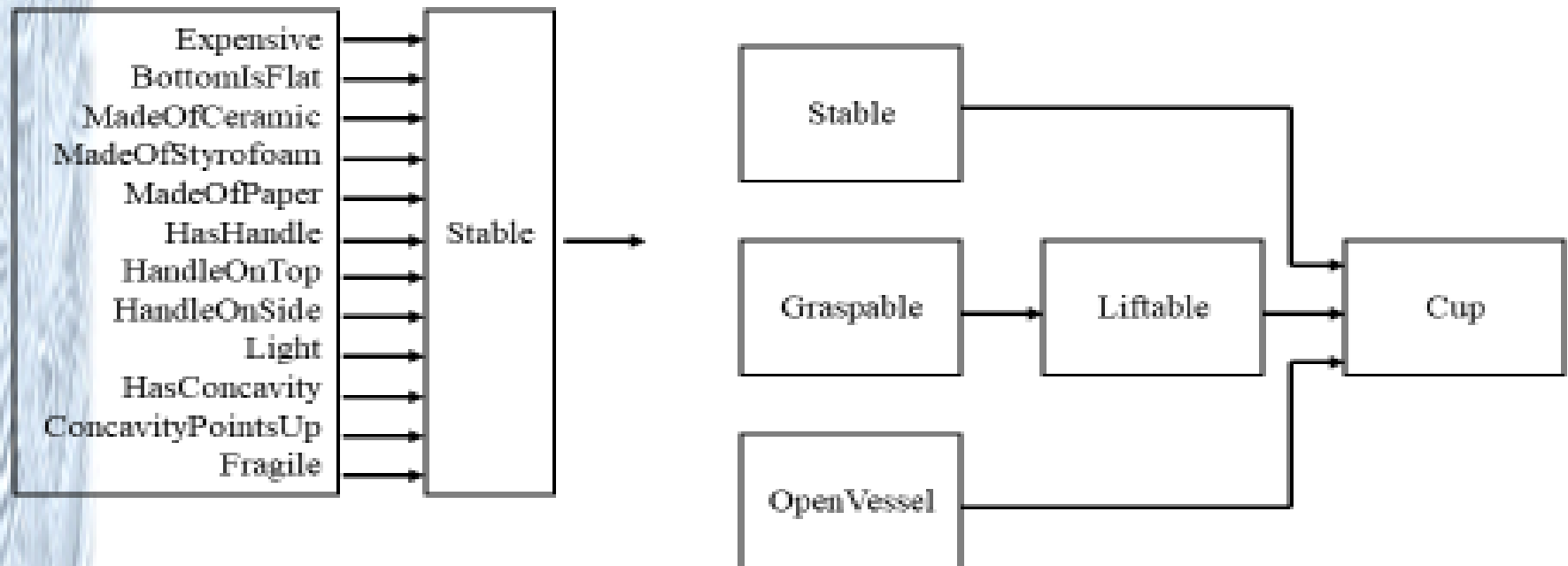


Training Examples

	Cups				Non-Cups			
BottomIsFlat	✓	✓	✓	✓	✓	✓	✓	✓
ConcavityPointsUp	✓	✓	✓	✓	✓		✓	✓
Expensive	✓		✓				✓	✓
Fragile	✓	✓			✓	✓	✓	✓
HandleOnTop					✓		✓	
HandleOnSide	✓			✓				✓
HasConcavity	✓	✓	✓	✓	✓		✓	✓
HasHandle	✓			✓	✓		✓	✓
Light	✓	✓	✓	✓	✓	✓	✓	✓
MadeOfCeramic	✓				✓		✓	
MadeOfPaper				✓				✓
MadeOfStyroForm		✓	✓		✓			✓

Explanation in Terms of Domain Theory

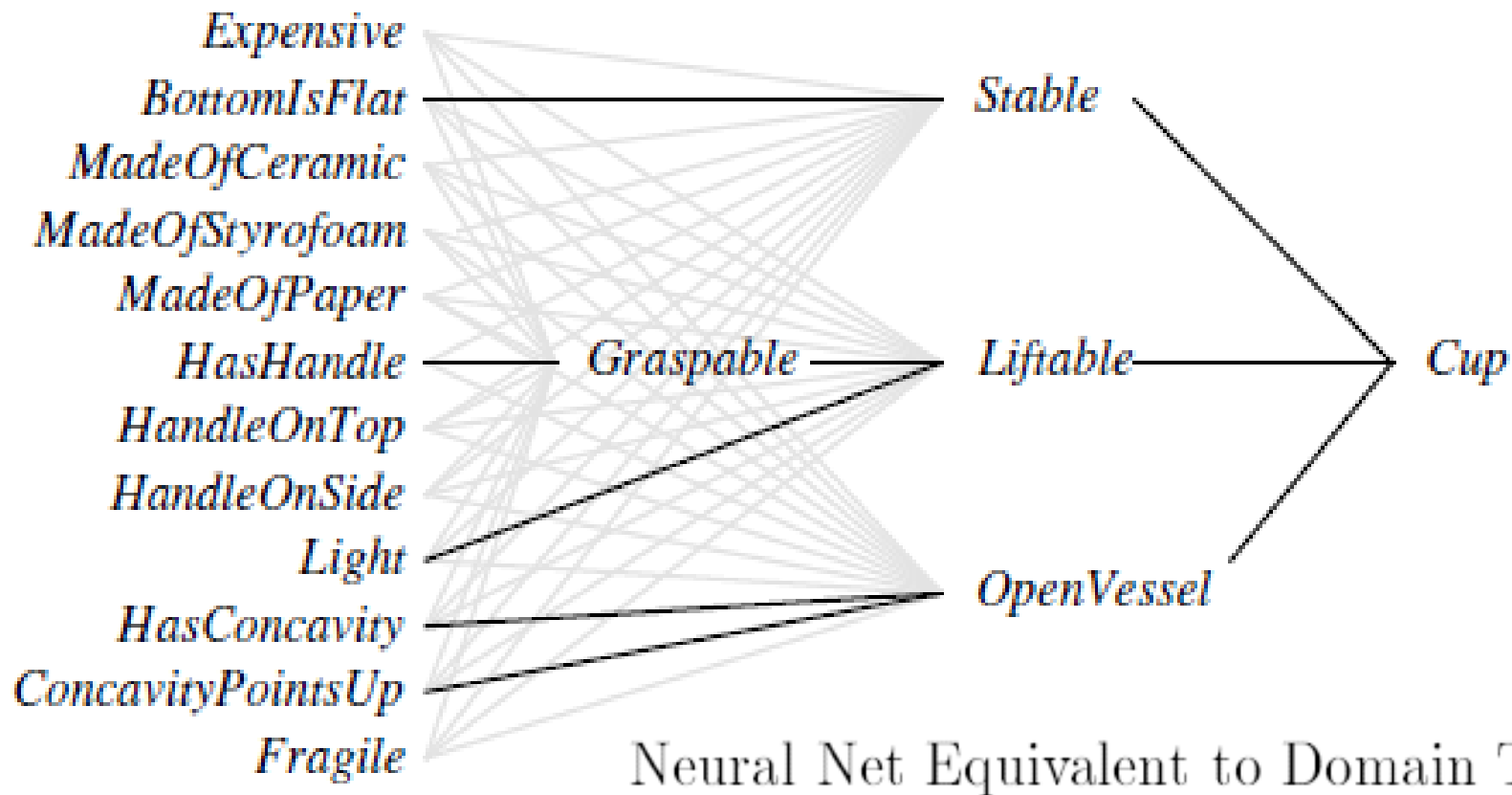
Prior learned networks for useful concepts combined into a single target network



KBANN (data D , domain theory B)

1. Create a feedforward network h equivalent to B
2. Use BACKPROP to tune h to fit D

- KBANN is an hybrid learning system (combining both domain knowledge + Training examples)



Creating Network Equivalent to Domain Theory

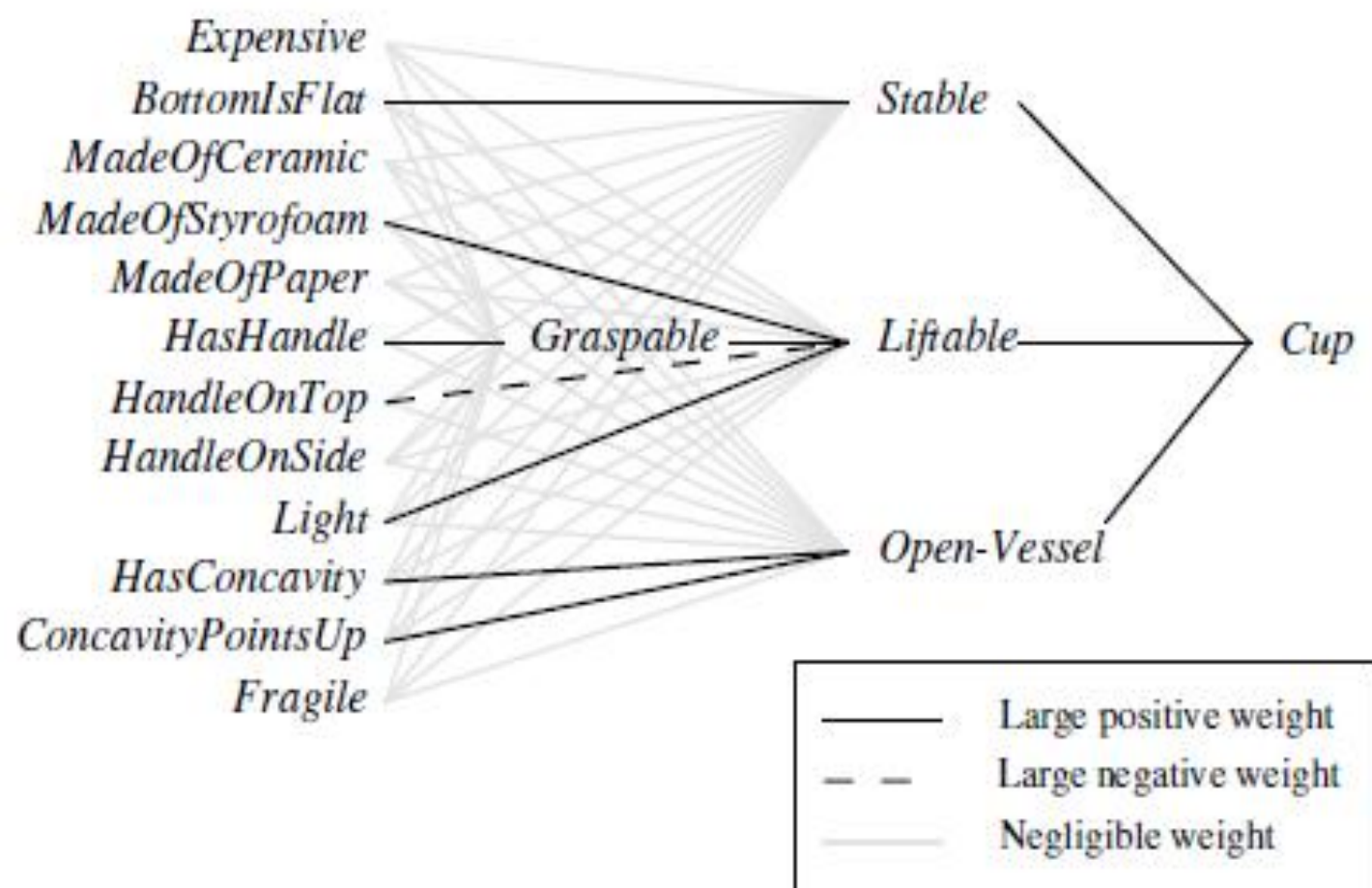
Create one unit per horn clause rule (i.e., an AND unit)

- Connect unit inputs to corresponding clause antecedents
- For each non-negated antecedent, corresponding input weight $w \leftarrow W$, where W is some constant
- For each negated antecedent, input weight $w \leftarrow -W$
- Threshold weight $w_0 \leftarrow -(n - .5)W$, where n is number of non-negated antecedents

Finally, add many additional connections with near-zero weights

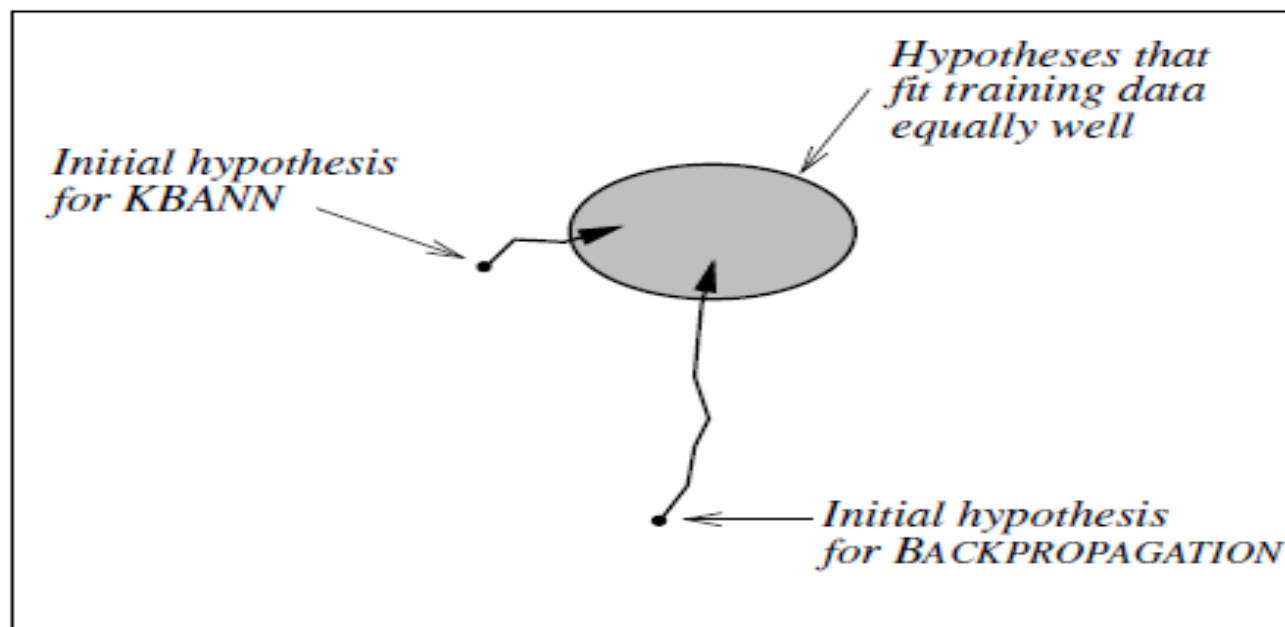
$$Liftable \leftarrow Graspable, \neg Heavy$$

Result of refining the network



Hypothesis space search in KBANN

Hypothesis Space



THANK YOU